MGMT 682: Digital Product Design
Assignment 2: Platforms and Pricing

Please write your answers as clearly and concisely as possible. Please type your answers. Recall the course policy on AI: you are free to use AI tools subject to disclosure when and how it was used, and you are fully responsible for everything you turn in.

You are working on a platform business that brings together consumers and advertisers. Assume we are working for an online content platform that adopts the advertising business model. You can assume that there are zero marginal costs to deliver the content.

Our business brings together consumers (C) and advertisers (A). Our data science team has done some analysis to estimate the demand curve of both users and advertisers, and our job is to find the optimal price for this offering.

Consumer demand curve does not depend on the advertiser:

$$q_C(p_C) = 100 - 1.1 \times p_C$$

Advertiser demand curve depends on the user base

$$q_A(p_C, p_A) = \gamma \times q_C(p_C, p_A) - p_A$$

I have uploaded an Excel spreadsheet to help solve this problem. **The worksheet can help you solve the problem, but please write your answers below.** The cells in red are the inputs to this problem. Fix $\gamma = 1$ at first.

1. Build out the spreadsheet. Create formulas that, as a function of prices, calculate the quantity sold to each side of the market (yellow cells) and profits received in each side of the market (blue cells)

There is nothing to turn in for this question; it is only meant to help you solve the remaining problems.

Given:
We run a two-sided content platform with consumers (C) and advertisers (A). Marginal costs are zero.

Consumer demand (independent of advertisers):
qC(pC) = 100 − 1.1 pC

Advertiser demand (depends on user base):
qA(pC, pA) = γ qC(pC) − pA

We fix γ = 1, so: qA = qC − pA.
Profits (zero marginal cost):
πC = pC * qC, πA = pA * qA, πT = πC + πA

2. **Find the optimal price for each side of the market, treating the other side as fixed. We will solve this problem in steps.**
   a. **Find the consumer price ($\tilde{p}_C$) that maximizes profits on the consumer side of the market. Use the Excel solver tool to help here. (10 points)**

      Consumer-side profit:
      πC(pC) = pC(100 − 1.1 pC) = 100 pC − 1.1 pC^2

      Differentiate and set to zero:
      d(πC)/d(pC) = 100 − 2.2 pC = 0
      pC^(Ind) = 100/2.2 = 500/11 = 45.45
      Answer 2(a): pC^(Ind) = 500/11 = 45.45

   b. **Calculate the quantity of consumers this price attracts $q_c(\tilde{p}_C)$. (10 points)**

      Quantity of consumers at pC^(Ind):

      qC^(Ind) = 100 − 1.1 pC^(Ind) = 100 − 1.1*(500/11) = 100 − 50 = 50.

      Answer 2(b): qC^(Ind) = 50.

   c. **Find the advertiser price ($\tilde{p}_A$) that maximizes the profits on the advertiser side of the market, given the quantity of consumers that we receive at the consumer price $\tilde{p}_C$. Use the Excel solver tool again here. (10 points)**

      pA that maximizes advertiser-side profit given qC^(Ind):

      Given qC^(Ind) = 50: qA = 50 − pA.

      Advertiser profit: πA(pA) = pA(50 − pA) = 50 pA − pA^2.

      Differentiate and set to zero: d(πA)/d(pA) = 50 − 2 pA = 0.

      pA^(Ind) = 25.

      Answer 2(c): pA^(Ind) = 25.

d. Calculate the profits we attain at the prices $\tilde{p}_C$, $\tilde{p}_A$ (10 points)

Profits at independent prices:

Advertiser quantity: qA^(Ind) = qC^(Ind) − pA^(Ind) = 50 − 25 = 25.

Consumer profit: πC^(Ind) = pC^(Ind) * qC^(Ind) = (500/11)*50 = 25000/11, about 2272.73.

Advertiser profit: πA^(Ind) = 25*25 = 625.

Total profit: πT^(Ind) = 2272.73 + 625 = about 2897.73.

Answer 2(d): πC^(Ind) = 25000/11 (about 2272.73), πA^(Ind) = 625, πT^(Ind) about 2897.73.

3. Next, we will jointly choose the profits to maximize total profits and see how the prices and profits change.

a. Use the excel solver tool to jointly choose p_C^*, p_A^* to maximize total profits. (10 points) (10 points)

Maximize total profit πT:

Total profit: πT = pC qC + pA(qC − pA).

Substitute qC = 100 − 1.1 pC:

πT(pC, pA) = pC(100 − 1.1 pC) + pA[(100 − 1.1 pC) − pA].

First-order conditions:

1) With respect to pA: ∂πT/∂pA = (100 − 1.1 pC) − 2 pA = 0, so pA = (100 − 1.1 pC)/2.

2) With respect to pC: ∂πT/∂pC = 100 − 2.2 pC − 1.1 pA = 0.

Solving the two equations gives pC^(J) = 9000/319 (about 28.21) and pA^(J) = 1000/29 (about 34.48).

Answer 3(a): pC^(J) = 9000/319 (about 28.21), pA^(J) = 1000/29 (about 34.48).

b. Calculate the quantities and profits we attain at these prices (10 points)

Quantities and profits at joint prices:

Consumers: qC^(J) = 100 − 1.1 pC^(J) = 2000/29, about 68.97.

Advertisers: qA^(J) = qC^(J) − pA^(J) = (2000/29) − (1000/29) = 1000/29, about 34.48.

Consumer profit: πC^(J) = pC^(J) * qC^(J), about 1945.74.

Advertiser profit: πA^(J) = pA^(J) * qA^(J), about 1189.06.

Total profit: πT^(J) = 1945.74 + 1189.06, about 3134.80.

Answer 3(b): qC^(J) = 2000/29 (about 68.97), qA^(J) = 1000/29 (about 34.48), πC^(J) about 1945.74, πA^(J) about 1189.06, πT^(J) about 3134.80.

4. Briefly explain the intuition for why prices and profits changed from problem (2) to (3). Why did one price go down and the other up? (10 points)

Answer:
In independent pricing, each side is optimized in isolation, so cross-side benefits are not considered. Under joint pricing, the platform internalizes the fact that a larger consumer base increases advertiser demand and value. Hence, it lowers the consumer price to expand participation and raises the advertiser price because access to a bigger audience is more valuable. This aligns with the rule of subsidizing the value-creating side and charging the side that benefits from it.

5. Calculate the user surplus under both the independent pricing and joint pricing schemes. Explain the intuition for why they are different. Give a real-world example where consumer surplus increases due to platform pricing. (15 points)

Consumer surplus under independent vs joint pricing

Inverse consumer demand comes from qC = 100 − 1.1 pC, so pC(q) = (100 − q)/1.1.

Consumer choke price (qC = 0): pMaxC = 100/1.1 = 1000/11, about 90.91.

Consumer surplus for linear demand: CS_C = 0.5 * (pMaxC − pC) * qC.

Independent pricing: CS_C^(Ind) = 0.5 * (90.91 − 45.45) * 50 = about 1136.36.

Joint pricing: CS_C^(J) = 0.5 * (90.91 − 28.21) * 68.97 = about 2161.93.

Intuition: joint pricing lowers pC to expand qC, so consumers gain more surplus. Real-world example: Google Search, Instagram, and YouTube charge users zero and monetize on advertisers.

6. Calculate the advertiser surplus under both the independent pricing and joint pricing schemes. Explain the intuition for why these differ. (15 points)

Advertiser surplus under independent vs joint pricing

With $\gamma = 1$, qA = qC − pA. Inverse advertiser demand is pA(qA) = qC − qA, so advertiser choke price is pMaxA = qC.

Advertiser surplus: CS_A = 0.5 * (pMaxA − pA) * qA = 0.5 * (qC − pA) * qA.

Independent pricing:

qC^(Ind) = 50, pA^(Ind) = 25, qA^(Ind) = 25.

CS_A^(Ind) = 0.5 * (50 − 25) * 25 = 312.5.

Joint pricing:

qC^(J) is about 68.97, pA^(J) is about 34.48, qA^(J) is about 34.48.

CS_A^(J) = 0.5 * (68.97 − 34.48) * 34.48 = about 594.53.

Conclusion: Joint pricing increases qC, shifting advertiser demand outward. Even with a higher pA, advertisers buy more and gain more surplus.

AI Disclosure:

I used AI to learn the concepts taught in the class before proceeding with this assignment, and to help with grammar and clarity.